AI CLINICAL DECISION SUPPORT LITE · HACKATHON 2026

Evidence-Based Supplement Assistant

Answers sports-nutrition questions from one official NIH fact sheet — and cites the page every claim came from.

```
Loading is 20 g/day of
creatine monohydrate in
four portions of 5 g
[1, p. 19].
```

↑ *the page is checked against the index, never taken from the model*

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THE PROBLEM

Most people who train take supplements. Almost none can check them.

66%

of surveyed US college students take a supplement

[1, p. 1]

42%

of ~21,000 college athletes take protein products

[1, p. 1]

2 in 3

elite track and field athletes, across international surveys

[1, p. 1]

$5.7bn

of sports nutrition sold in a single year

[1, p. 1]

Every figure above was retrieved from the document this system indexes, by this system — which is the product, demonstrated on itself.

The evidence is real, and unreadable

The answers exist — in a 32-page federal fact sheet written for health professionals, dense with study designs, dose ranges and hedged conclusions.

The market is louder than the evidence

Products are sold as blends while research tests single ingredients, and labels rarely state how much of each is inside.

Even the experts hedge

The National Athletic Trainers' Association notes that study outcomes are often equivocal, which makes these substances controversial and confusing to use [1, p. 2].

WHY A LANGUAGE MODEL ALONE IS NOT ENOUGH

It will answer. It will not show you where the answer came from.

1. **It sounds exactly like evidence**

   A dose, a loading protocol, a timing window — delivered in the register of a clinical guideline, with nothing behind it.

2. **It accepts a false premise**

   Ask “since DHEA raises testosterone, what dose should athletes take?” and it supplies one. The fact sheet says DHEA does not improve performance at all.

3. **It cannot be audited**

   A coach or dietitian cannot check a claim they cannot trace. Without a document, a section and a page, review is impossible.

What retrieval changes

The model stops being the source of facts and becomes the thing that phrases them.

1. **Closed corpus**

   It may use the retrieved passages and nothing else — no outside knowledge, no inference past what the text states.

2. **Provenance it cannot fake**

   Every claim carries [n, p. X], and the page is looked up from the index after generation. Choosing the source is the model's job; the page is not.

3. **A refusal that means something**

   When retrieval turns up nothing bearing on the question, the answer is that the document does not cover it.

ARCHITECTURE

Four layers, one class per stage, and an evaluation loop around all of it

PDF › Parse + structure › Chunk + page tag › Embed › Vector index › Retrieve › Rerank › Ground + cite › Answer

1 · Ingestion

A PDF carries no headings, only glyphs at coordinates. Font sizes above the body size are ranked into heading levels, recovering 95 sections; each page's text is tagged so every chunk knows its page.

2 · Retrieval

The query is expanded into other phrasings, searched against 1,144 rows — passages plus questions generated for them — then reordered by a reranker before the cut.

3 · Generation

Grounding is strict, and the citation the model writes is rewritten against the index, so the page comes from retrieval metadata rather than from the model.

4 · Safety & evaluation

Refusal when nothing relevant is retrieved, plus a 34-question golden set scored on four metrics — two grading retrieval, two grading the answer.

Everything runs on local models through one OpenAI-shaped client, so no user's health question leaves the machine and swapping providers is a base URL and two model ids.

THE RETRIEVAL PROBLEM · 1

Retrieve a little, and the evidence never reaches the answer

We started where most systems start: dense search, take the top five passages, answer from those.

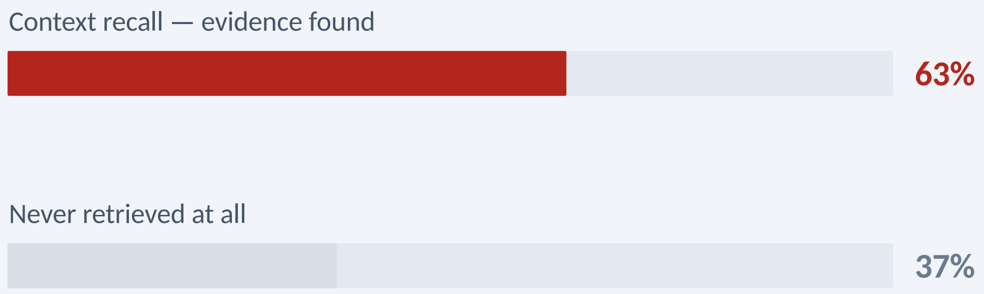
THE PROBLEM

About a third of the needed evidence never arrives

Context recall sat in the low sixties — the retriever was finding some of what each answer needed and missing the rest. The failures were not random. Comparative questions, multi-part questions and one misspelled query were the ones that broke, and the worst of them retrieved nothing useful at all.

A passage that is never retrieved cannot be cited, cannot be checked, and cannot appear in the answer. This is the ceiling on everything downstream.

Where the evidence went



Measured across 34 questions written to stress the retriever, not to flatter it.

Retrieve more, and the answer gets diluted instead

So we looked deeper — twice as many passages per question. Recall rose sharply, and a second problem took its place.

THE NEW PROBLEM

The evidence arrives, buried in what came with it

Recall climbed into the mid-eighties and every multi-part and comparative question was answered — including the misspelled one, which turned out never to have been a spelling problem. The right passages had been found all along and ranked below the cut.

But precision fell into the low seventies: most of what now reached the answerer was off-topic. Answers grew longer, hedged more, and drifted from the question asked. Depth bought recall and paid for it in relevance.

Shallow → deeper

a real trade, not a free win

	Shallow		Deeper
Context recall	63%	→	85%
Context precision	79%	→	74%
Answer relevancy	77%	→	77%

Depth cannot fix ranking. It only moves the cut further down a list that is still in the wrong order.

Retrieve deep, then reorder — and keep both

Before building anything we measured where the evidence actually sits. Two fifths of it was being retrieved and then ranked too low: a reordering problem, not a recall problem.

THE FIX

Fetch twenty, have the model order them, keep ten

The embedder scores a question and a passage separately — neither vector ever sees the other. A reranker reads them together, which is why it can tell a passage that shares vocabulary from one that actually answers.

Listwise, so a search costs one extra call rather than one per candidate. It reorders and never filters: anything the model leaves out keeps its original place behind what it chose, and a failed call falls back to the dense order.

Measured against the same setup without it

Context recall	85% →	82%	−3
Context precision	74% →	82%	+8
Answer relevancy	77% →	78%	+1

The cost is real and we are not hiding it: reranking decides which ten of the twenty reach the answer, so it can drop one dense had kept. Context recall and faithfulness each give up about three points to buy eight of context precision.

WHAT WE CAN PROVE

Four metrics, 34 questions, and the one we still fail

87%

Faithfulness

of the answer's claims are supported by the passages shown

PASS

82%

Context precision

of the passages retrieved were relevant to the question

PASS

82%

Context recall

of the evidence each answer needed was found

PASS

78%

Answer relevancy

how directly the answer addresses the question asked

BELOW 80%

SUCCESS CASE

“HMB or BCAAs for recovery?” — the reranker pulls the BCAA passages from ranks 13, 15 and 18 into the top ten, so both halves of the comparison reach the answer, each cited to its page.

REFUSAL CASE

Three questions the document cannot answer, plus a false-premise DHEA dose and a value the page never states. All five are declined rather than invented — and the metric scores a correct refusal as a failure, which is why we report it.

THE HONEST LIMIT

Answer relevancy is short of threshold. Part of it is the judge sampling fewer generated questions than it asks for; part of it is genuinely longer answers than the question needs.

Scored by an LLM judge over a golden set built to break the system: near-miss pairs, multi-hop questions, a negation, a misspelling, and three questions with no answer in the document at all.

WHAT COMES NEXT

From “what does the evidence say” to “what should I eat”

The fact sheet already states doses. The next step is to meet them from food and drink, not only from a tub.

1

Match the dose, not just the ingredient

The document gives quantities — around 2 cups of beetroot juice, 3 g/day of HMB, a caffeine range per kilogram of body weight. Those are the numbers a recommendation has to hit.

2

Map ingredients onto foods and drinks

Beetroot juice, tart cherry, oily fish, dairy protein, coffee. A food composition source joins to the same ingredient names the index already carries, so provenance survives the join.

3

Answer the question people actually ask

“I am a 70 kg cyclist, what should I drink before a time trial?” becomes a portion and a timing, cited to the page that set the dose — and a refusal when the evidence does not support one.

The constraint stays the same. A food recommendation is only as good as the dose behind it, and the dose has to come from the document with its page attached. Nothing gets recommended that the evidence does not carry — which is the same rule that governs every answer the system gives today.

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